

- Let $x \in A \cap B$. By Defn. 5, $x \notin A \cap B$.
- By Defn. 2, $\neg(x \in A) \wedge (x \in B)$ is True.
- By 1st De Morgan's Law of propositions, $\neg(x \in A) \vee \neg(x \in B)$ is True.
- By Defn. of OR, Atleast one of $\neg(x \in A)$ and $\neg(x \in B)$ is True.
- $\therefore$ By Defn. of negation of proposition, atleast one of $x \notin A$ and $x \notin B$ is True.
- By Defn. 5, atleast one of $x \in \bar{A}$ and $x \in \bar{B}$ is True.
- WLOG, let $x \in \bar{A}$. By Defn. of OR, $x \in \bar{A} \vee x \in \bar{B}$ is True.
- By Defn. 1, $x \in \bar{A} \cup \bar{B}$. x is any arbitrary element of $A \cap B$.
- By Defn. 3 of Section 2.1, $A \cap B \subseteq \bar{A} \cup \bar{B}$.
- Let $x \in \bar{A} \cup \bar{B}$. By Defn. 1, $x \in \bar{A} \vee x \in \bar{B}$ is True.
- By Double-negation Law of Propositions, $\neg(\neg(x \in \bar{A} \vee x \in \bar{B}))$ is True.
- By 2nd De Morgan's Law of Propositions, $\neg(\neg(x \in \bar{A}) \wedge \neg(x \in \bar{B}))$ is True.
- By Defn. 5, $\neg(\neg(x \notin A) \wedge \neg(x \notin B))$ is True.
- By Defn. of negation of proposition, $\neg(x \in A \wedge x \in B)$ is True.
- By Defn. 2, $\neg(x \in A \cap B)$ is True.
- By Defn. of negation of proposition, $x \notin A \cap B$ is True. By Defn. 5, $x \in \overline{A \cap B}$ is True.
- Since x is an arbitrary element of $\bar{A} \cup \bar{B}$, By Defn. 3 of Sec-2.1, $\bar{A} \cup \bar{B} \subseteq \overline{A \cap B}$. $\therefore \bar{A} \cup \bar{B} = \overline{A \cap B}$ $\square$

Example 11: Use set builder notation and logical equivalences to establish the first De Morgan law $\overline{A \cap B} = \bar{A} \cup \bar{B}$.

Solution: We can prove this identity with the following steps.

$$\begin{aligned}
 \overline{A \cap B} &= \{x \mid x \notin A \cap B\} && \text{by definition of complement} \\
 &= \{x \mid \neg(x \in (A \cap B))\} && \text{by definition of does not belong symbol} \\
 &= \{x \mid \neg(x \in A \wedge x \in B)\} && \text{by definition of intersection} \\
 &= \{x \mid \neg(x \in A) \vee \neg(x \in B)\} && \text{by the first De Morgan Law for logical equivalences}
 \end{aligned}$$